

Wireless Link Layer, MAC protocols, CSMA

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Wireless Link Layer

- Link layer (layer 2) is above physical layer
 - Gives frames to PHY to transmit
 - Gets correct (and incorrect) frames from PHY
- Link layer has a notion of a “link” between sender and receiver, adds a link layer header to identify src / dst
- Link layer functions
 - **Medium access control (MAC)**: share access to the broadcast wireless medium
 - **Link rate adaptation**: configure suitable PHY parameters (modulation / coding rate) to match the bit rate and signal quality of each link
 - **Error control**: retransmissions to recover from errors

MAC protocols

- MAC protocols arbitrate access to the medium
- Two broad classes
 - **Scheduling-based / centralized-** A central entity decides who sends when. E.g., cellular base stations schedule transmissions of users on uplink and downlink.
 - **Contention-based / decentralized-** Nodes decide who transmits when in a distributed fashion. E.g., WiFi clients and access point contend for medium using CSMA MAC protocol.

The idea of multiplexing

- MAC protocols **multiplex** users over time, frequency, etc.
 - **Time division multiplexing (TDM)** – different users transmit at different times. The time of transmission can be decided by a central entity (scheduling-based) or in a distributed fashion (contention-based)
 - **Frequency division multiplexing (FDM)** – different users use different frequencies. Frequency allocation typically happens in a centralized fashion.
 - Some **combination of the above**. For example, different sets of users are allocated different frequencies (FDM) followed by multiple users in the same frequency band sharing channel using TDM.
 - Other dimensions –space division multiplexing (use different geographic regions), code division multiplexing. These concepts will be clearer in the next lecture.
- If multiplexing is not done properly (e.g., two users send at same time on same frequency), the users interfere with each other.
- **Efficient multiplexing without causing interference** to any user is the goal of medium access control (MAC) protocols.

MAC protocols - Roadmap

- This lecture: **contention-based** MAC protocols
 - **ALOHA**: simplest such protocol
 - **CSMA**: widely used in WiFi etc
- Next lecture: **scheduling-based** MAC protocols
 - TDMA, and its variants
 - Different scheduling algorithms used in cellular networks
 - CDMA
 - OFDMA etc.

Contention-based MAC: ALOHA

- ALOHA is the simplest contention-based MAC protocol.
- N nodes in a network, all want to send packets on a shared medium.
- Only one node can successfully send at a time, more than one transmission leads to collision.
- In ALOHA, each node with a packet sends with probability p .
- Probability of successful transmission = probability that only one of the N nodes sends = $Np(1-p)^{(N-1)}$
- This probability is maximum when $p = 1/N$
- That is, ALOHA works efficiently when nodes choose the attempt probability p intelligently
- Even with this optimal choice of p , the fraction of time when useful data is sent (without collisions) for large N is roughly 18% (i.e., works out to $1/e$ mathematically). Not very efficient!
- Key idea – randomness is essential for distributed MAC

Can we do better?

- Yes we can. We can “listen” to the medium for some time to see if someone is sending. Transmit only if the medium is idle. This is the idea behind carrier sense.
- Ethernet (most common wired MAC) and the MAC layer of WiFi are both based on this idea.
- What is the difference between wired and wireless? In wired, the voltage levels across a wire do not change much between sender and receiver. So we know exactly when a collision happens. In wireless, the conditions at sender and receiver may be different, so cannot always detect collisions.
- Wireless – use link layer ACK to check if collision or not
- Ethernet implements CSMA/CD (carrier sense multiple access with collision detection)
- WiFi implements CSMA/CA (carrier sense multiple access with collision avoidance) since we cannot be fully sure of collisions.

Ethernet CSMA/CD: Prelude to 802.11

CSMA/CA (DCF)

- CSMA/CD: Carrier-Sense Multiple Access with Collision Detection
 - Listen before transmit (CS)
 - Tx when (as soon as) medium is free (1-persistent)
 - Collision Detection (CD)
 - Backoff (exponential) on collision

802.11 CSMA/CA

- Collision detection (near) impossible in wireless
 - ♦ Tx power is relatively very high near the transmitter
- Conceptual name is CSMA/CA: Carrier-Sense Multiple Access with Collision Avoidance
- 802.11 calls it DCF (Distributed Coordination Function)
- Collision Avoidance:
 - Backoff before tx (even when no collision)
 - Contention Window (CW) in terms of number of slots

CSMA (Carrier Sense Multiple Access)

- Time is divided into slots (a few microsec each). Nodes measure energy of channel over a slot to decide if channel is busy or idle
- Every node maintains a variable called its contention window (CW)
- When a node has a packet to send:
 - If channel is free, then send
 - If channel is busy, pick a “backoff counter” between 0 and CW
 - Wait for “r” idle slots. That is, in every slot where channel is idle, decrement backoff counter.
 - When backoff counter is zero, and channel is idle, start transmission in that slot
 - If no ACK, retransmit packet again (pick backoff counter, wait etc)

CSMA – contention window CW

$$T_{\text{backoff}} = \text{Random}(\text{CW}) \times \text{SlotTime}$$

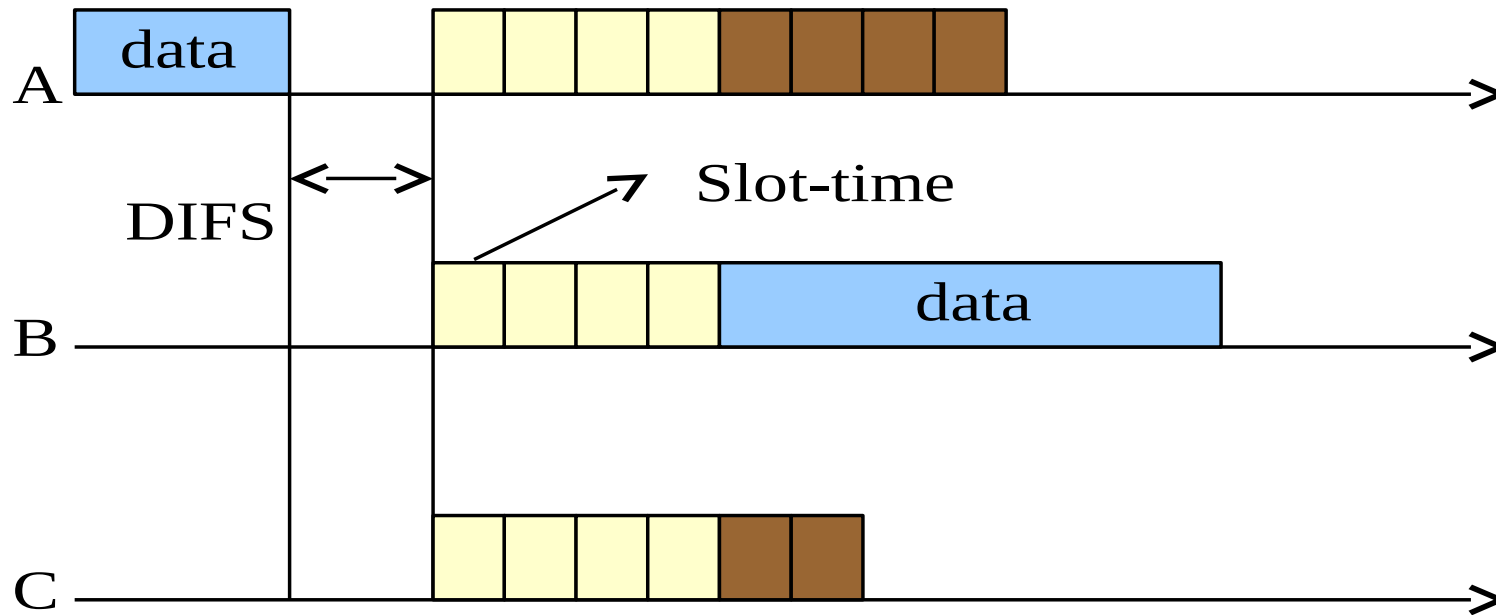
- CW decides how aggressively nodes contend for the channel. Roughly equivalent to the attempt probability in ALOHA.
- Initially, nodes use $\text{CW} = \text{CW_min}$
- When collision, **double CW** (up to CW_max). So successive retransmissions of a packet will see longer and longer backoffs
- When packet successfully sent after retries, reset **CW to CW_min**

CSMA – contention window CW

- If many nodes in the network, CW will be high on average
- Small CW – nodes will contend more as average backoff will be lower.
Large CW – contention is less

The Backoff Procedure

Use immediate ACK; no ACK $\Rightarrow$ assume collision



Contention window:

$\text{Numslots} = \text{Random}[0, \text{CW})$

$\text{CW} = 16, 32, 64, \dots, 1024$

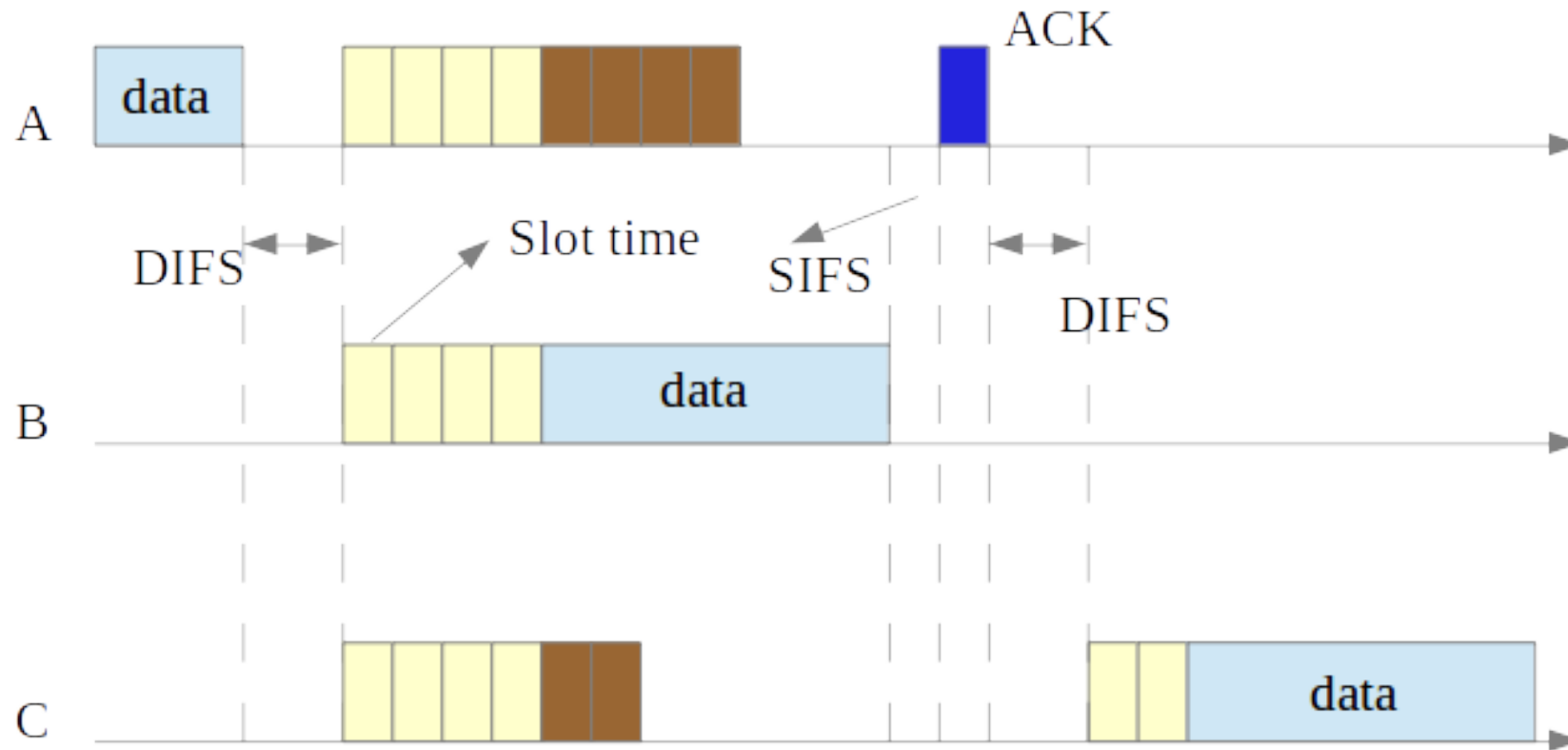
Double CW on collision

Question: Why DIFS?

CSMA – inter-frame spacing

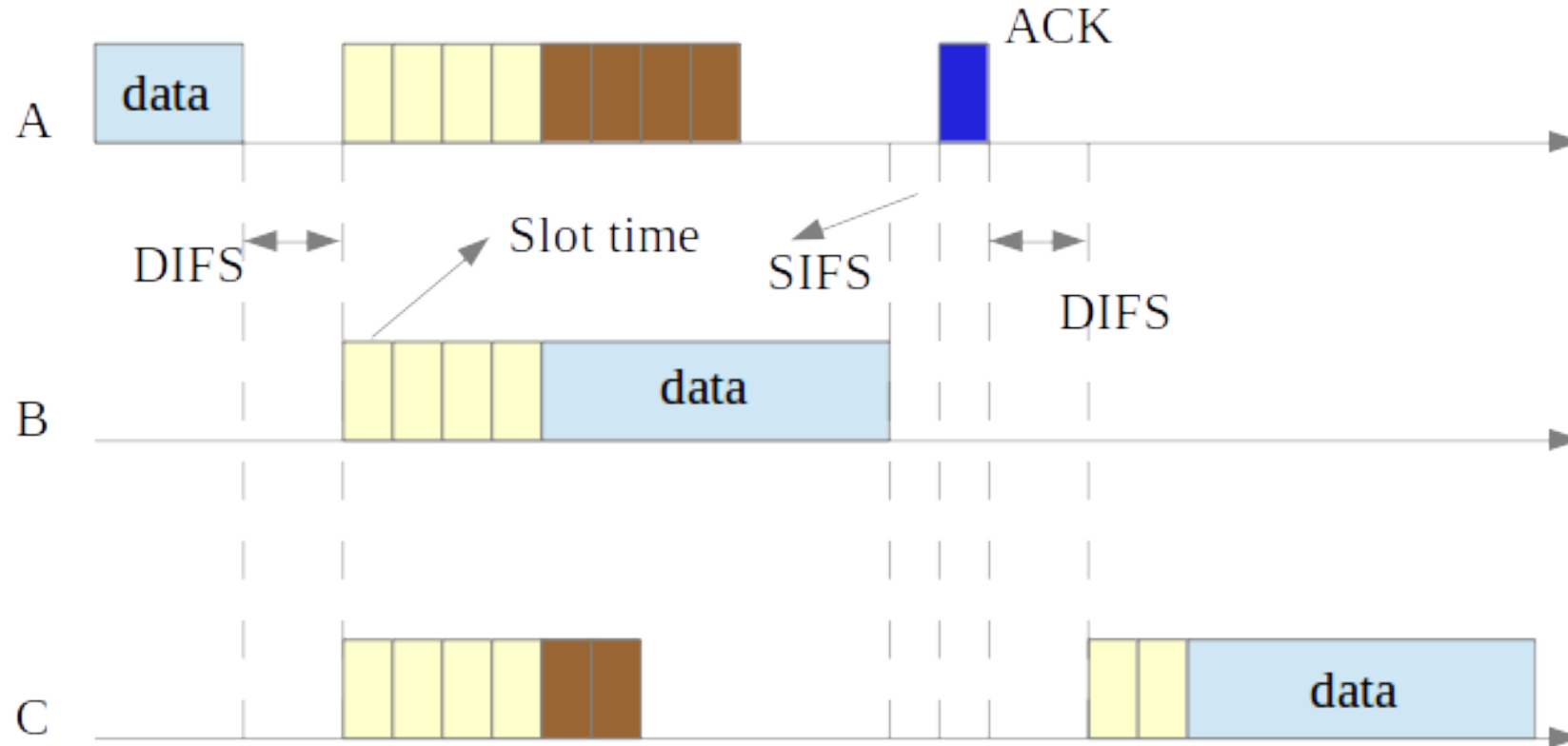
- After every data frame, the receiver sends an ACK if frame is successfully received
- ACK is sent immediately after data packet (after a “short inter frame spacing” SIFS)
- All other nodes wait for a longer inter frame spacing (DIFS) after a transmission. Allows sender and receiver to reset their PHY and get ready for a new transmission.
- That is, a transmission blocks the medium for the duration of the transmission + SIFS + duration of ACK + DIFS. Only then can other nodes contend

CSMA/CA + ACK



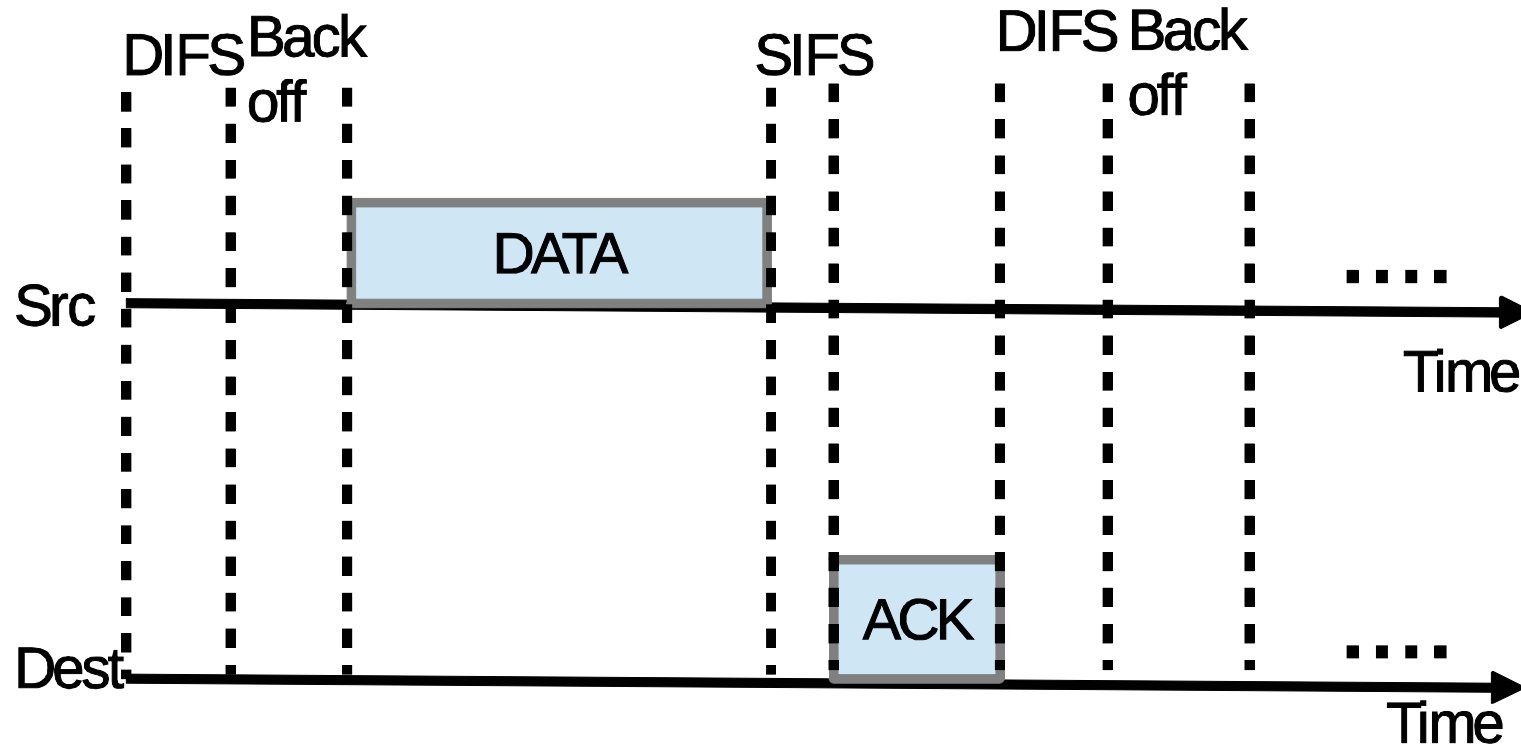
- ACK missing $\Rightarrow$ deduce collision, retransmit
→ Have to contend anew
- Question: Relation between SIFS and DIFS?

CSMA/CA + ACK

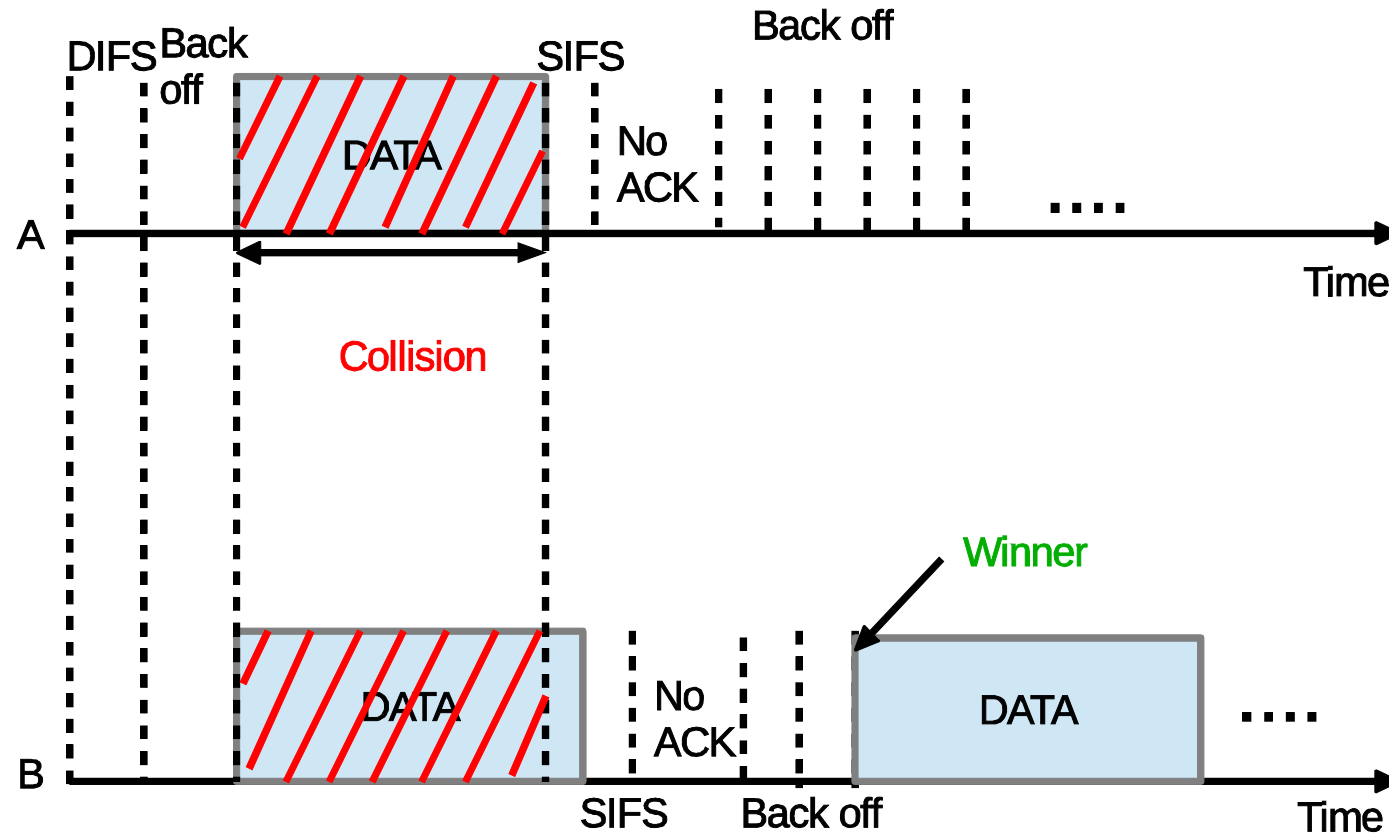


- ACK missing $\Rightarrow$ deduce collision, retransmit
 - $\rightarrow$ Have to contend anew
- Question: Relation between SIFS and DIFS?
 - $\rightarrow$ SIFS should be $<$ DIFS
 - $\rightarrow$ Else, ACK timeout may occur unnecessarily

CSMA (Carrier Sense Multiple Access)



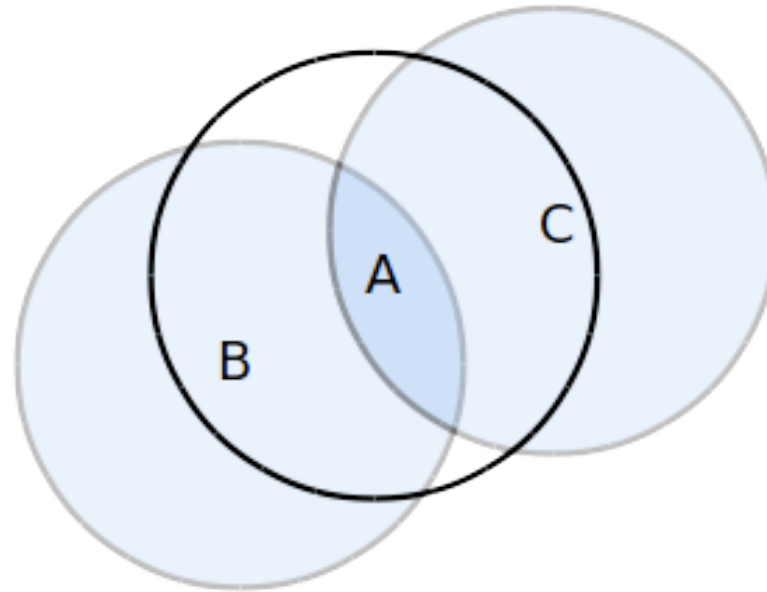
CSMA Collision



Problems with CSMA

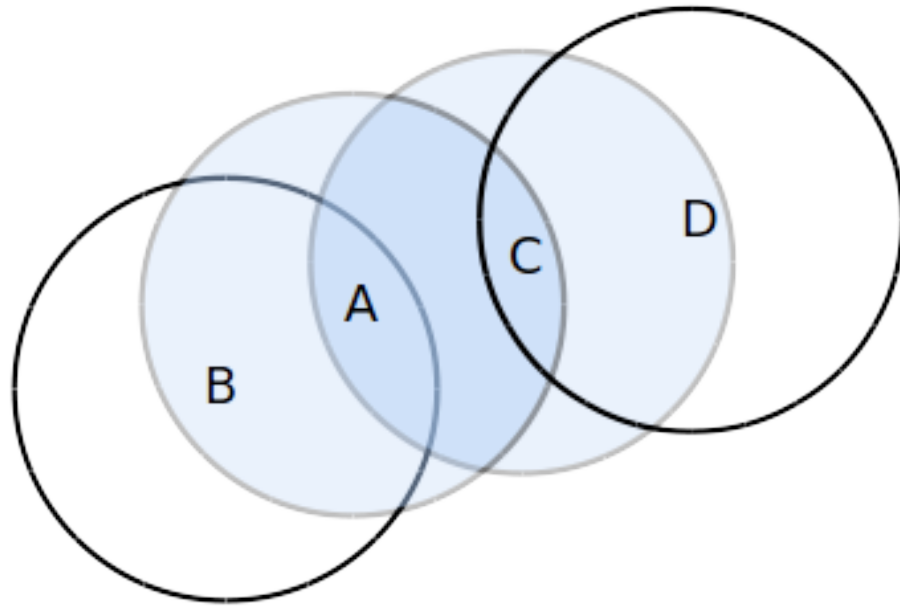
- Hidden node problem: $B \rightarrow A \leftarrow C$
 - B and C are out of carrier sense range of each other, so transmit simultaneously to A. Causes collisions at A.
- Exposed node problem: $B \leftarrow A \quad C \rightarrow D$
 - A and C are sending to far away receivers. They can both send concurrently, but do not do so due to carrier sense.
 - These problems because carrier sense does not tell you conditions at receiver, only at sender.

The Hidden Node Problem



Medium is free **DOES NOT IMPLY** ok to transmit

The Exposed Node Problem

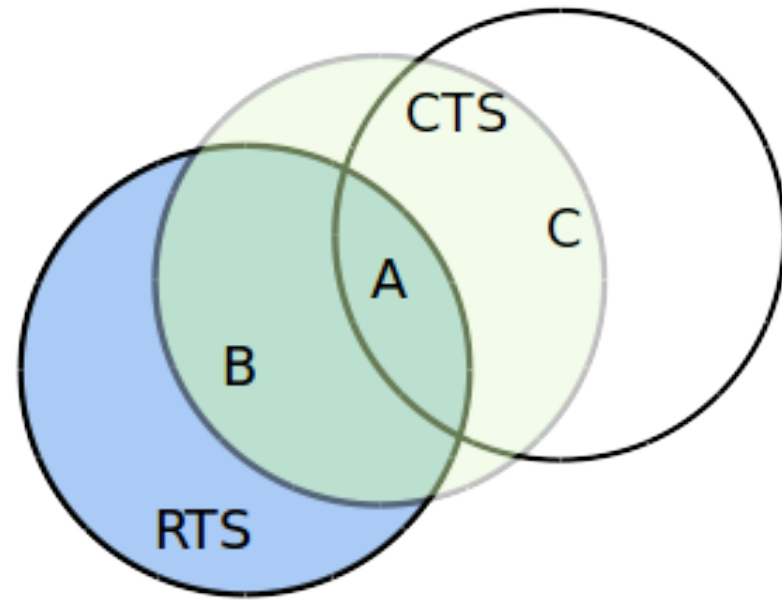


Medium is busy **DOES NOT IMPLY** not-ok to transmit

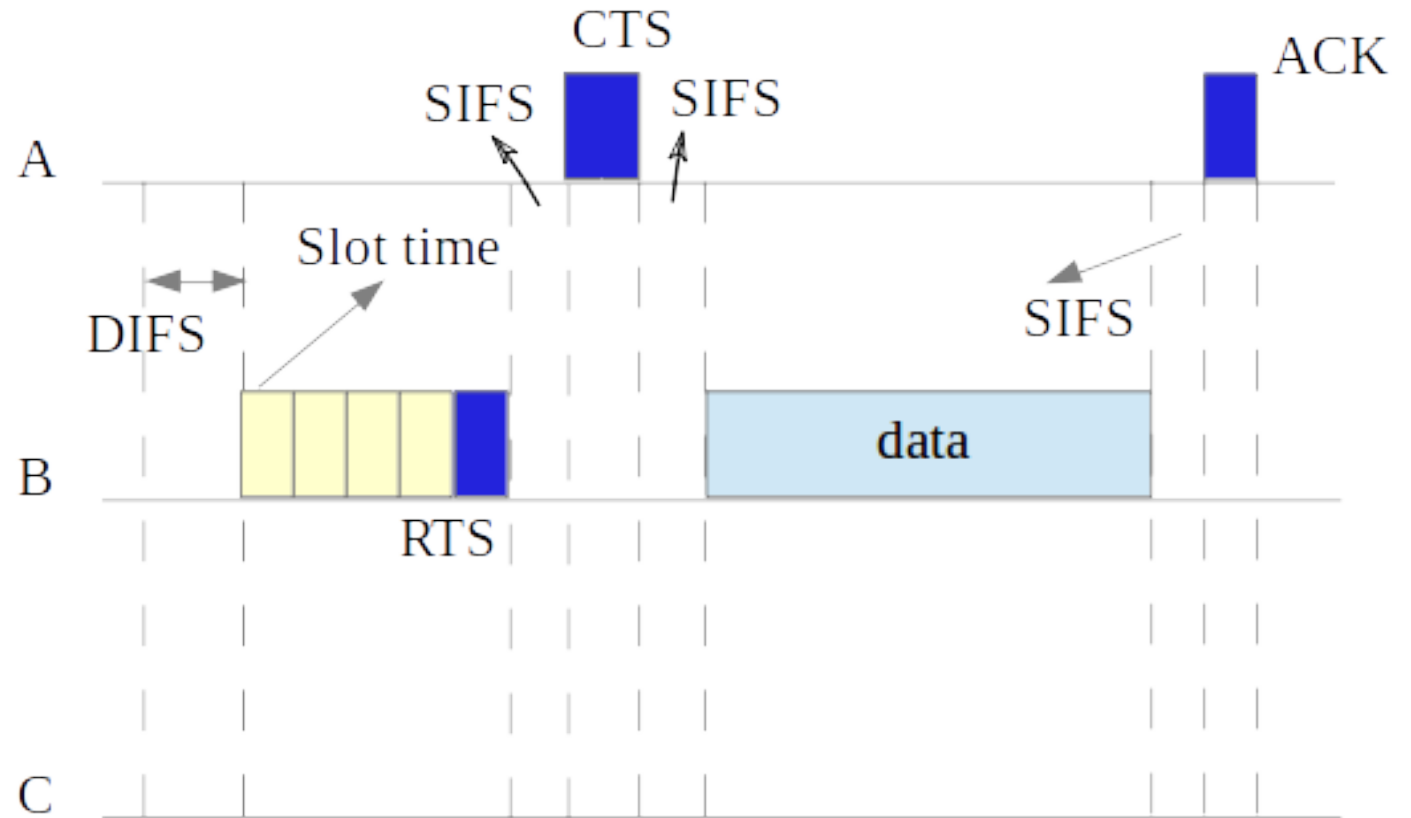
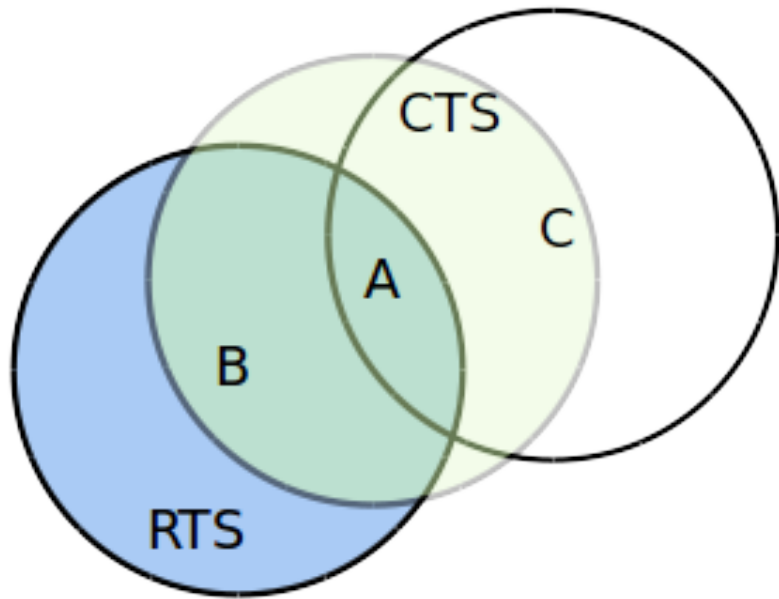
Problems with CSMA

- Fix: RTS/CTS mechanism. Before transmitting, B sends RTS (Request to Send). A sends CTS (Clear to Send). B sends data only after getting CTS. C hears CTS and avoids transmissions.

The Hidden Node Solution: RTS/CTS



The Hidden Node Solution: RTS/CTS



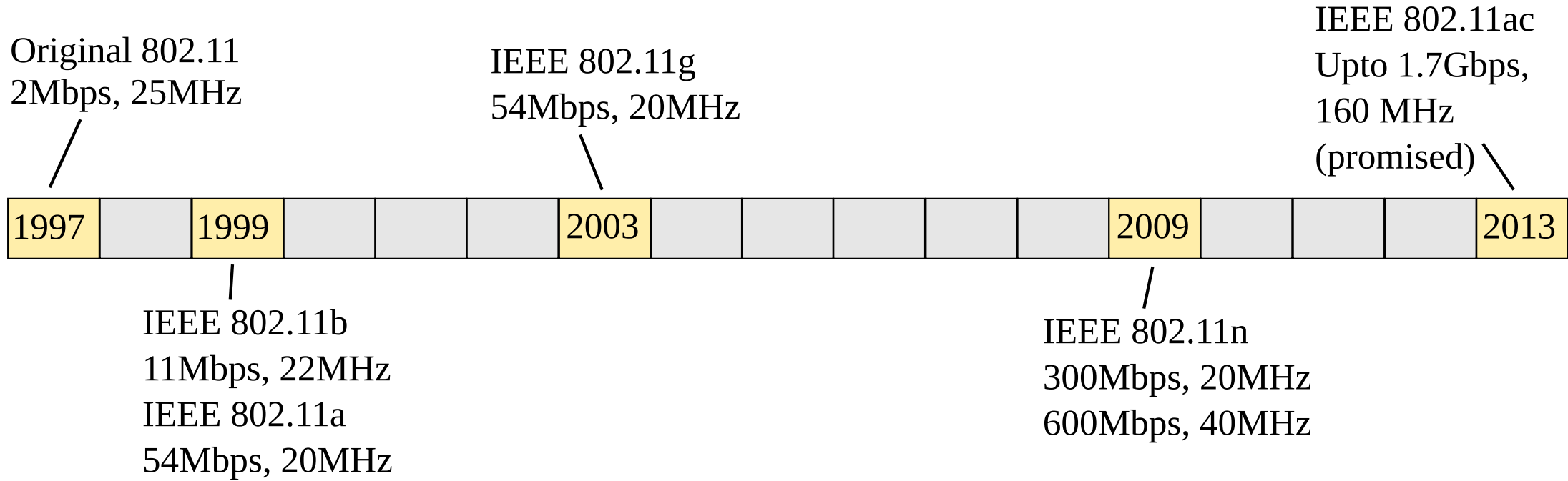
Problems with CSMA(2)

- RTS/CTS imposes high delay, especially in high throughput networks. Turned off by default.
- These problems because carrier sense does not tell you conditions at receiver, only at sender.

Problems with CSMA (2)

- Unfairness to high rate users. CSMA gives equal transmission opportunities. A node sending at 10Mbps gets 10 times *less* time than a node sending at 1 Mbps. The network is limited by the slowest node.
- When high user density, nodes often pick same backoff value and collide

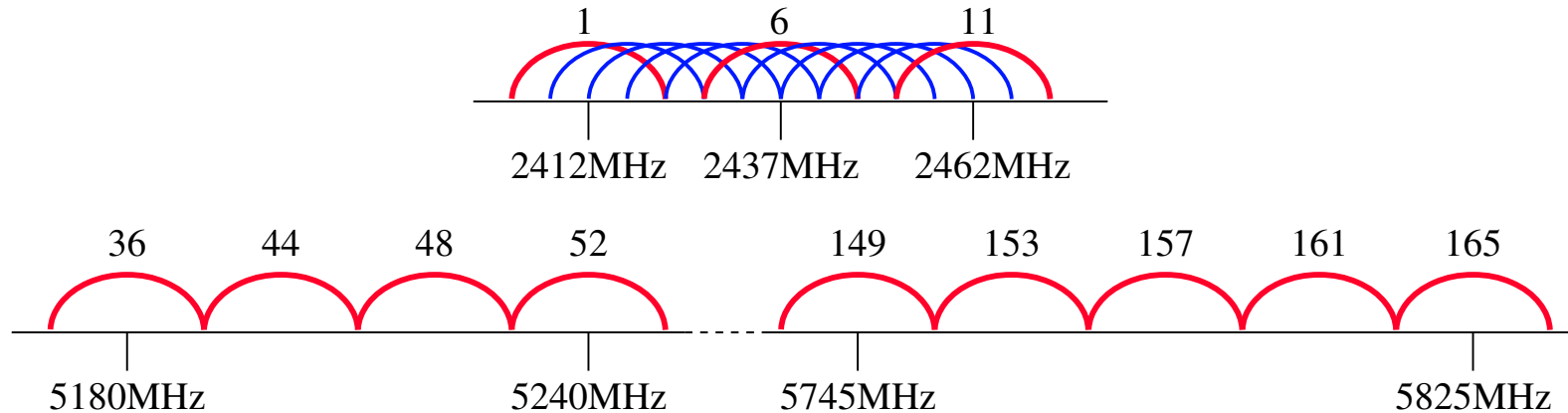
802.11



802.11: What does it Specify?

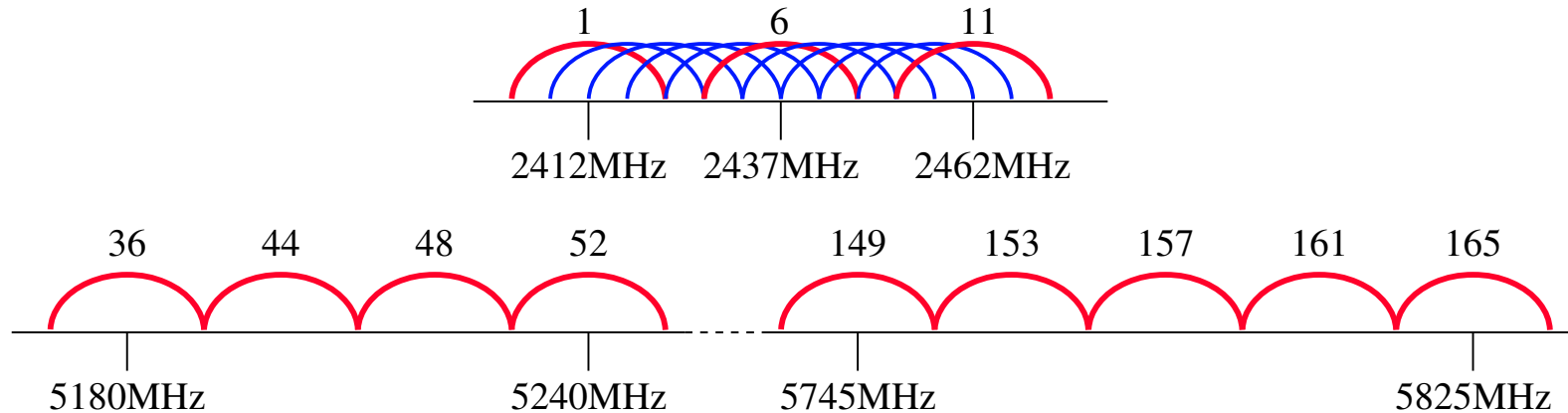
- **PHY** sub-layer
 - 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac
 - Modulation and coding schemes
- **MAC** sub-layer
 - Independent of the PHY
 - DCF (Distributed Coordination Function) based on CSMA/CA
 - PCF (Point Coordination Function): centralized control
 - **Question**: Which of DCF and PCF is implemented, and why?
- **Link** sub-layer
 - Framing, addressing, fragmentation, aggregation
- **Network management**
 - Network formation and maintenance

802.11 PHY Layer: Channels



- Designed to operate in ISM bands
- License free in most of the world, with transmit power restrictions
- **Question:** if license-free operation good or bad?

802.11 PHY Layer: Channels



- Designed to operate in ISM bands
- License free in most of the world, with transmit power restrictions
- **Question:** if license-free operation good or bad?
-
- Has been one of the keys to **WiFi's success**
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- Also cause of **performance problems!**